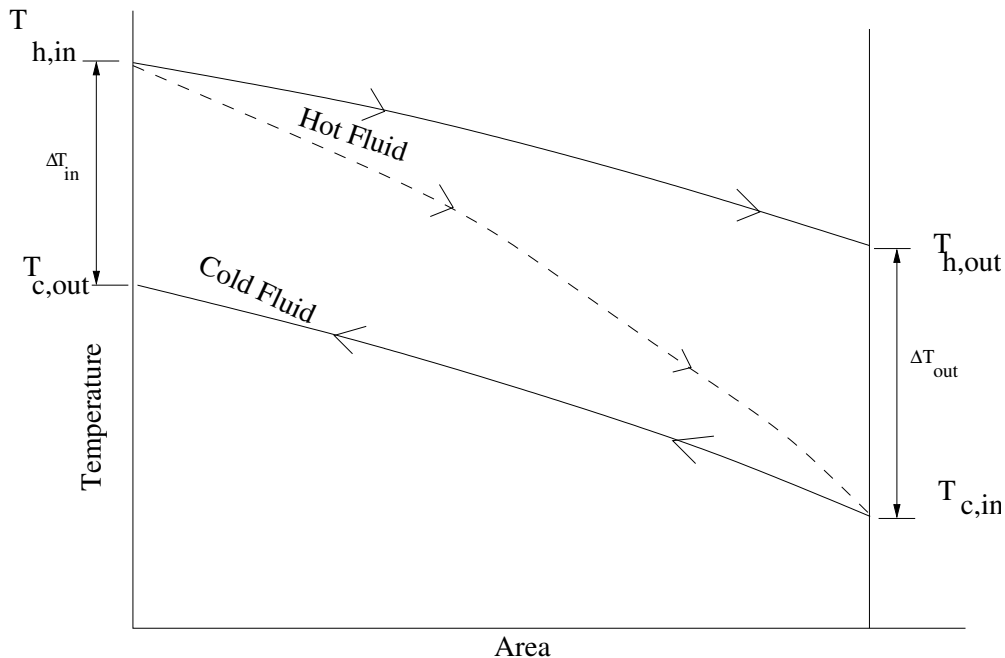


Effectiveness -NTU Method



- Heat Exchanger Sizing and Rating
- All Four Temperature of fluid may not be available
- Maximum temp. diff. possible $T_{h,out}$ can $\Rightarrow T_{c,in}$
- Heat capacity of fluids $m_h c_h = C_h$ and $m_c c_c = C_c$

- One heat capacity is minimum C_{min} and other maximum C_{max}
- Maximum heat transfer $C_{min} \times$ maximum temp diff.
- Maximum heat transfer $= C_{min}(T_{h,in} - T_{c,in})$
- Why C_{min} not C_{max} ?

- One heat capacity is minimum C_{min} and other maximum C_{max}
- Maximum heat transfer $C_{min} \times$ maximum temp diff.
- Maximum heat transfer $= C_{min}(T_{h,in} - T_{c,in})$
- Why C_{min} not C_{max} ?
- Will cause temperature crossing
- Effectiveness $\epsilon = \frac{\text{Actual Heat Transfer}}{\text{Maximum possible Heat Transfer}}$
- From Eq (1), We have $\ln \left(\frac{T_{h,out} - T_{c,out}}{T_{h,in} - T_{c,in}} \right) = -UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)$
- $\left(\frac{T_{h,out} - T_{c,out}}{T_{h,in} - T_{c,in}} \right) = e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)}$
- We have $C_h(T_{h,in} - T_{h,out}) = C_c(T_{c,out} - T_{c,in})$
- $T_{h,out} = T_{h,in} - \frac{C_c}{C_h}(T_{c,out} - T_{c,in})$
- Substituting $\left(\frac{T_{h,in} - \frac{C_c}{C_h}(T_{c,out} - T_{c,in}) - T_{c,out}}{T_{h,in} - T_{c,in}} \right) = e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)}$

- Re-arranging $\left(\frac{T_{h,in} - \frac{C_c}{C_h} T_{c,out} + \frac{C_c}{C_h} T_{c,in} - T_{c,out}}{T_{h,in} - T_{c,in}} \right) = e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)}$

- $\left(\frac{T_{h,in} - T_{c,in} + T_{c,in} - \frac{C_c}{C_h} T_{c,out} + \frac{C_c}{C_h} T_{c,in} - T_{c,out}}{T_{h,in} - T_{c,in}} \right) = e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)}$

- $\left(\frac{T_{h,in} - T_{c,in} + T_{c,in} \left(1 + \frac{C_c}{C_h} \right) - \left(1 + \frac{C_c}{C_h} \right) T_{c,out}}{T_{h,in} - T_{c,in}} \right) = e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)}$

-

$$1 - \left[\frac{T_{c,out} - T_{c,in}}{T_{h,in} - T_{c,in}} \right] \left(1 + \frac{C_c}{C_h} \right) = e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)} \quad (3)$$

- We have Effectiveness $\epsilon = \frac{\text{Actual Heat Transfer}}{\text{Maximum possible Heat Transfer}}$

- Actual heat transfer

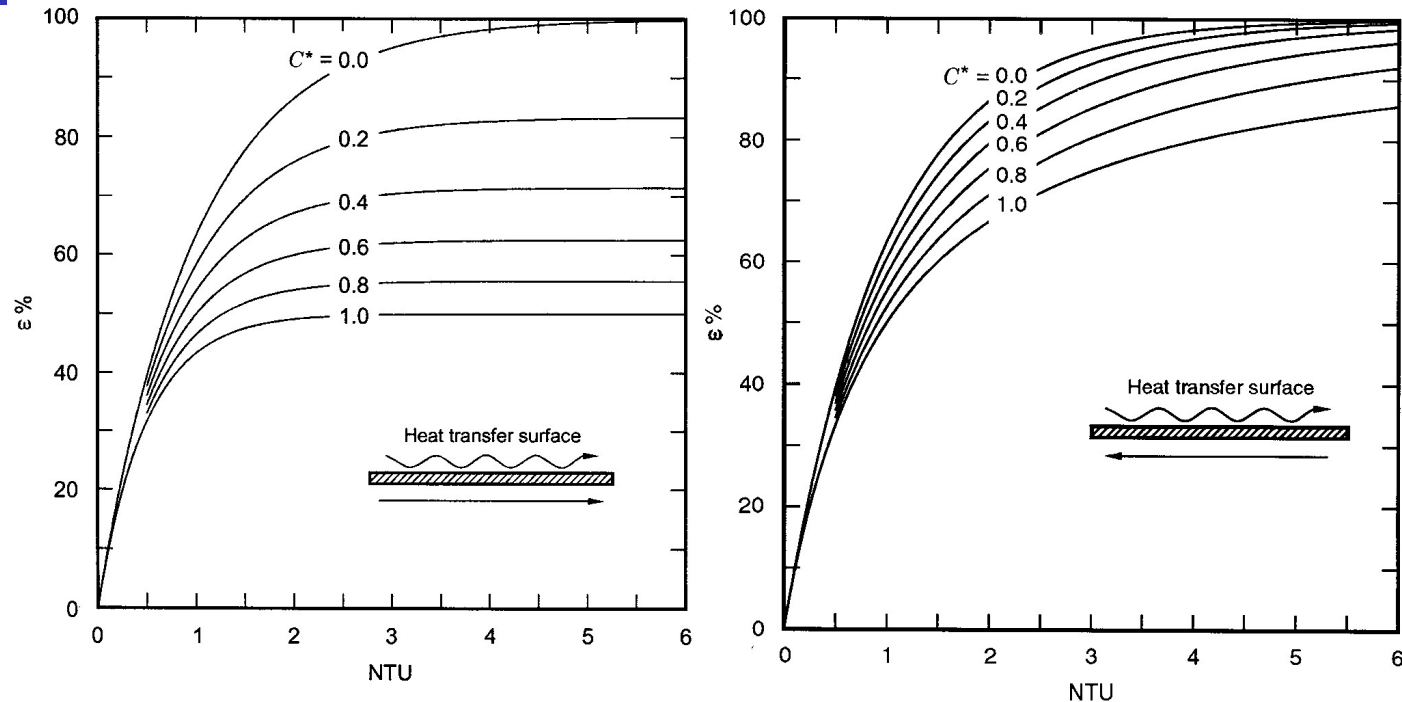
$$Q_{ac} = C_h(T_{h,in} - T_{h,out}) = C_c(T_{c,out} - T_{c,in})$$

- Maximum heat transfer $Q_{max} = C_{min}(T_{h,in} - T_{c,in})$

- Effectiveness $\epsilon = \frac{C_c(T_{c,out} - T_{c,in})}{C_{min}(T_{h,in} - T_{c,in})}$
- Re-arranging $(T_{c,out} - T_{c,in}) = \epsilon \frac{C_{min}}{C_c} (T_{h,in} - T_{c,in})$
- Substituting in Eq (3)
- $1 - \epsilon \frac{C_{min}}{C_c} \left(1 + \frac{C_c}{C_h}\right) = e^{-UA\left(\frac{1}{C_h} + \frac{1}{C_c}\right)}$
- $\epsilon = \frac{1 - e^{-UA\left(\frac{1}{C_h} + \frac{1}{C_c}\right)}}{\frac{C_{min}}{C_c} \left(1 + \frac{C_c}{C_h}\right)}$
- $\epsilon = \frac{1 - e^{-UA\left(\frac{1}{C_h} + \frac{1}{C_c}\right)}}{\left(\frac{C_{min}}{C_c} + \frac{C_{min}}{C_h}\right)}$

- $$\epsilon = \frac{1 - e^{-UA \left(\frac{1}{C_h} + \frac{1}{C_c} \right)}}{\left(\frac{C_{min}}{C_c} + \frac{C_{min}}{C_h} \right)}$$
- $$\epsilon = \frac{1 - e^{-\frac{UA}{C_{min}} \left(\frac{C_{min}}{C_h} + \frac{C_{min}}{C_c} \right)}}{\left(\frac{C_{min}}{C_c} + \frac{C_{min}}{C_h} \right)}$$
- We define Number of heat transfer units (NTU) $N = \frac{UA}{C_{min}}$
- $$\epsilon = \frac{1 - e^{-N \left(\frac{C_{min}}{C_h} + \frac{C_{min}}{C_c} \right)}}{\left(\frac{C_{min}}{C_c} + \frac{C_{min}}{C_h} \right)}$$
- We define $C = \frac{C_{min}}{C_{max}}$
- Out of C_h and C_c One will be maximum and the other will be minimum
- Thus for parallel flow, $\epsilon = \frac{1 - \exp[-N(1+C)]}{1+C}$

Effectiveness -NTU Method Rating problem



- Find NTU from U , Area and C_{min}
- From the graph corresponding to the type of HE find ϵ
- Using ϵ we can find the exit temperature

Heat Exchanger Design -Remarks

- Significance of NTU
- Difference between LMTD method and Effectiveness-NTU method
- Heat exchanger design considerations
- Pressure drop, materials
- Type of service
- Heat Exchanger optimization
- Heat Exchanger Design Softwares